



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Lipid metabolism Part III: lipid metabolism disorders

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**Harvey and Ferrier: Lippincott Biochemistry 8th Edition: Chapter 18, 25, and 26
Harper's Illustrated Biochemistry, 32e: Chapter 26**

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Learning Objectives

1. Categorize the different **hyper**lipoproteinemia and **hypo**lipoproteinemia.
2. Explain the different biochemical pathways that could potentially be targeted pharmacologically in the management of heart diseases.
3. Identify the **statins** as the main therapeutic intervention for treating **dys**lipidemia/atherosclerosis and explain the biochemical basis of their actions.
4. Illustrate the risk factors of the metabolic syndrome and its specific lipid abnormalities.

Disorders related to lipoproteins

Name	Defect	Remarks
Hypolipoproteinemias Abetalipoproteinemia	No chylomicrons, VLDL, or LDL are formed because of defect in the loading of apo B with lipid.	Rare; blood acylglycerols low; intestine and liver accumulate acylglycerols. Intestinal malabsorption. Early death avoidable by administration of large doses of fat-soluble vitamins, particularly vitamin E.
Familial α -lipoprotein deficiency Tangier disease Fish-eye disease Apo A-I deficiencies	All have low or near absence of HDL.	Tendency toward hypertriacylglycerolemia as a result of absence of apo C-II, causing inactive LPL. Low LDL levels. Atherosclerosis in the elderly.
Hyperlipoproteinemias Familial lipoprotein lipase deficiency (type I)	Hypertriacylglycerolemia due to deficiency of LPL, abnormal LPL, or apo C-II deficiency causing inactive LPL.	Slow clearance of chylomicrons and VLDL. Low levels of LDL and HDL. No increased risk of coronary disease.
Familial hypercholesterolemia (type IIa)	Defective LDL receptors or mutation in ligand region of apo B-100. Or mutation in PCSK9	Elevated LDL levels and hypercholesterolemia, resulting in atherosclerosis and coronary disease.
Familial type III hyperlipoproteinemia (broad β -disease, remnant removal disease, familial dysbetalipoproteinemia)	Deficiency in remnant clearance by the liver is due to abnormality in apo E. Patients lack isoforms E3 and E4 and have only E2, which does not react with the E receptor. ^a	Increase in chylomicron and VLDL remnants of density <1.019 (β -VLDL). Causes hypercholesterolemia, xanthomas, and atherosclerosis.
Familial hypertriacylglycerolemia (type IV)	Overproduction of VLDL often associated with glucose intolerance and hyperinsulinemia.	Cholesterol levels rise with the VLDL concentration. LDL and HDL tend to be subnormal. This type of pattern is commonly associated with coronary heart disease, type 2 diabetes mellitus, obesity, alcoholism, and administration of progestational hormones.
Familial hyperalphalipoproteinemia	Increased concentrations of HDL.	A rare condition apparently beneficial to health and longevity.
Hepatic lipase deficiency	Deficiency of the enzyme leads to accumulation of large triacylglycerol-rich HDL and VLDL remnants.	Patients have xanthomas and coronary heart disease.
Familial lecithin:cholesterol acyltransferase (LCAT) deficiency	Absence of LCAT leads to block in reverse cholesterol transport. HDL remains as nascent disks incapable of taking up and esterifying cholesterol.	Plasma concentrations of cholesteryl esters and lysolecithin are low. Present is an abnormal LDL fraction, lipoprotein X, found also in patients with cholestasis. VLDL is abnormal (β -VLDL).
Familial lipoprotein(a) excess	Lp(a) consists of 1 mol of LDL attached to 1 mol of apo(a). Apo(a) shows structural homologies to plasminogen.	Premature coronary heart disease due to atherosclerosis, plus thrombosis due to inhibition of fibrinolysis.

^aThere is an association between patients possessing the apo E4 allele and the incidence of Alzheimer disease. Apparently, apo E4 binds more avidly to β -amyloid

A general strategy to identify and treat a clinical condition

- Identify which gene is mutated
- Review the physiological **function**(s) of the protein encoded by the mutated gene
- Rationalize the **clinical manifestations** based on its physiological function(s)
- Select an actionable pharmacological **intervention** to compensate or correct the defect(s)

Hyperlipoproteinemia

Hyper-: higher than normal

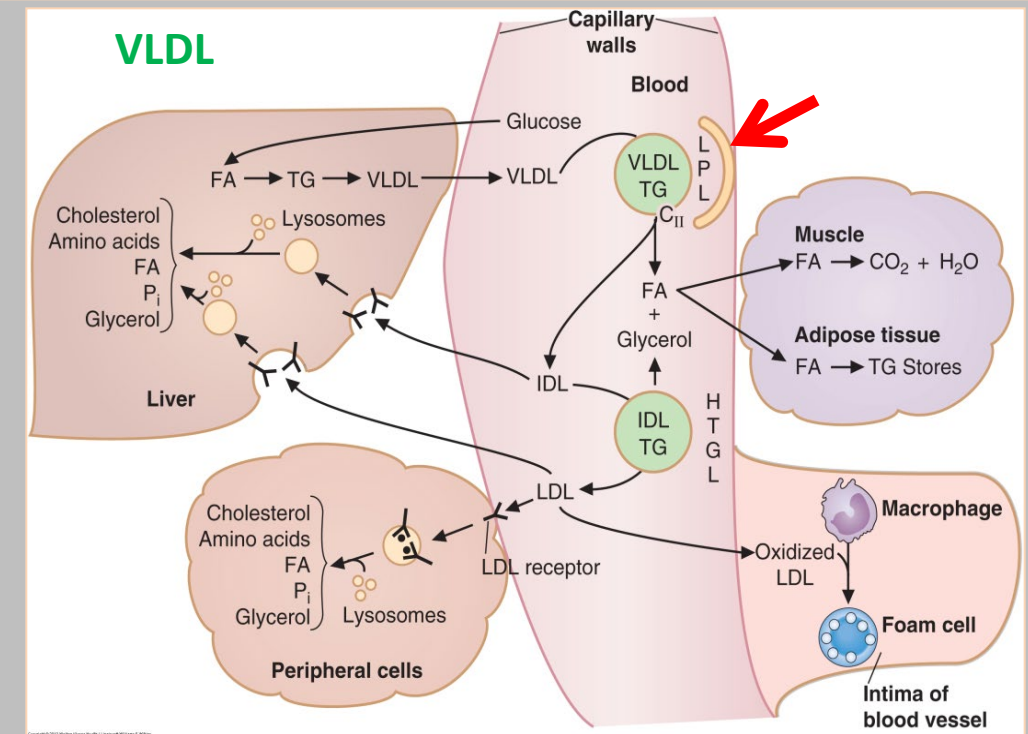
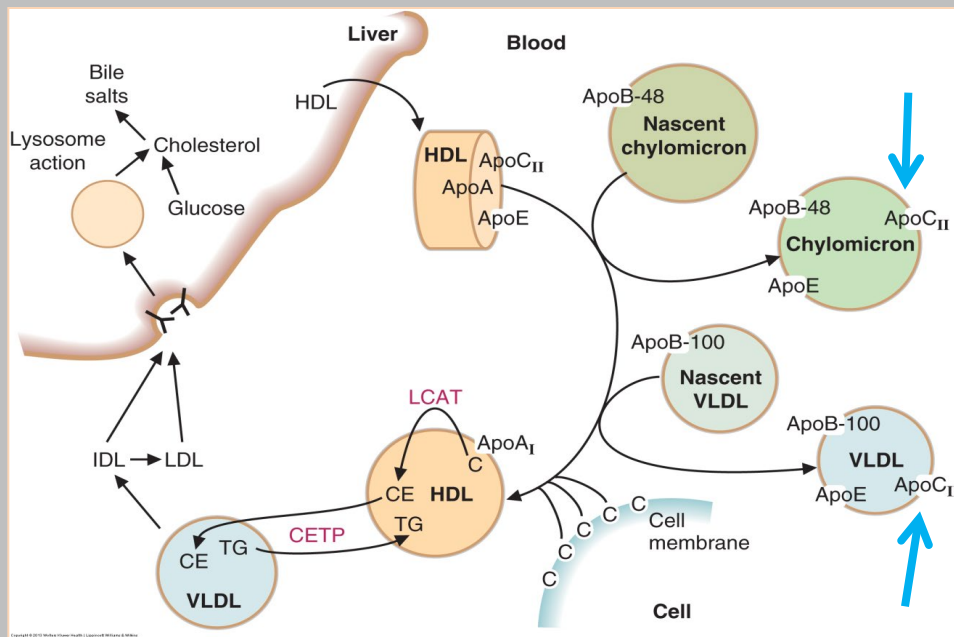
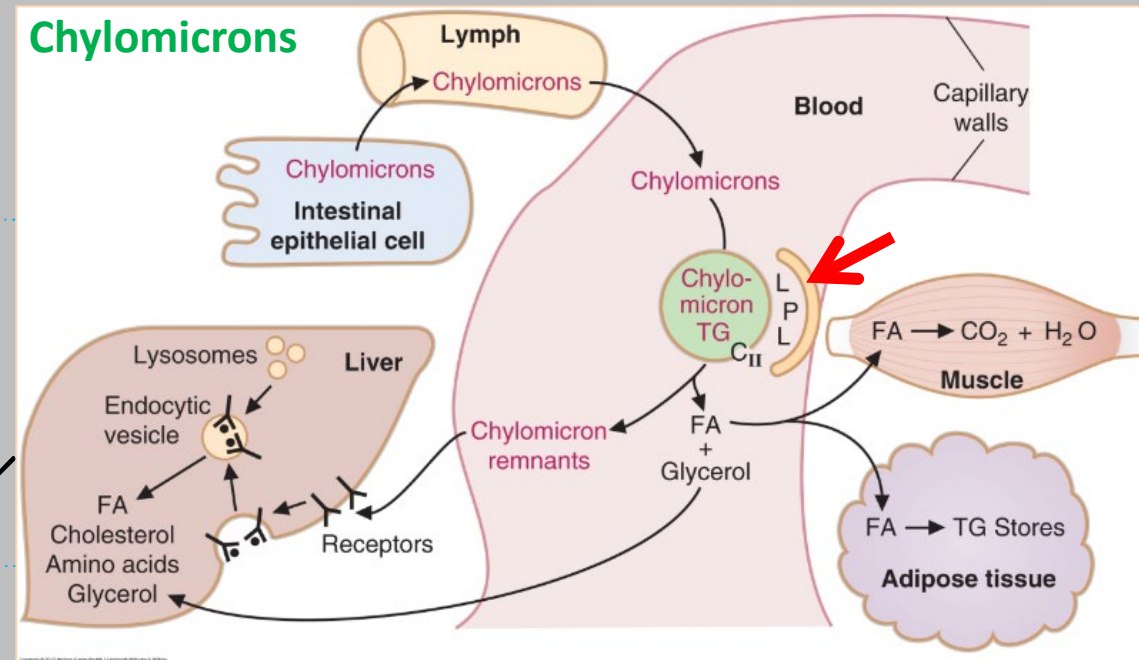
Hyperlipoproteinemia (Type I) – familial lipoprotein lipase (LPL) deficiency

Defective: **LPL** – digest triglyceride (TG)

ApoC-II – activate LPL

Consequences: chylomicron $\uparrow$, VLDL $\uparrow$, LDL $\downarrow$, HDL $\downarrow$

Increased risk of coronary disease: No



Hyperlipoproteinemia (Type IIa) – familial hypercholesterolemia

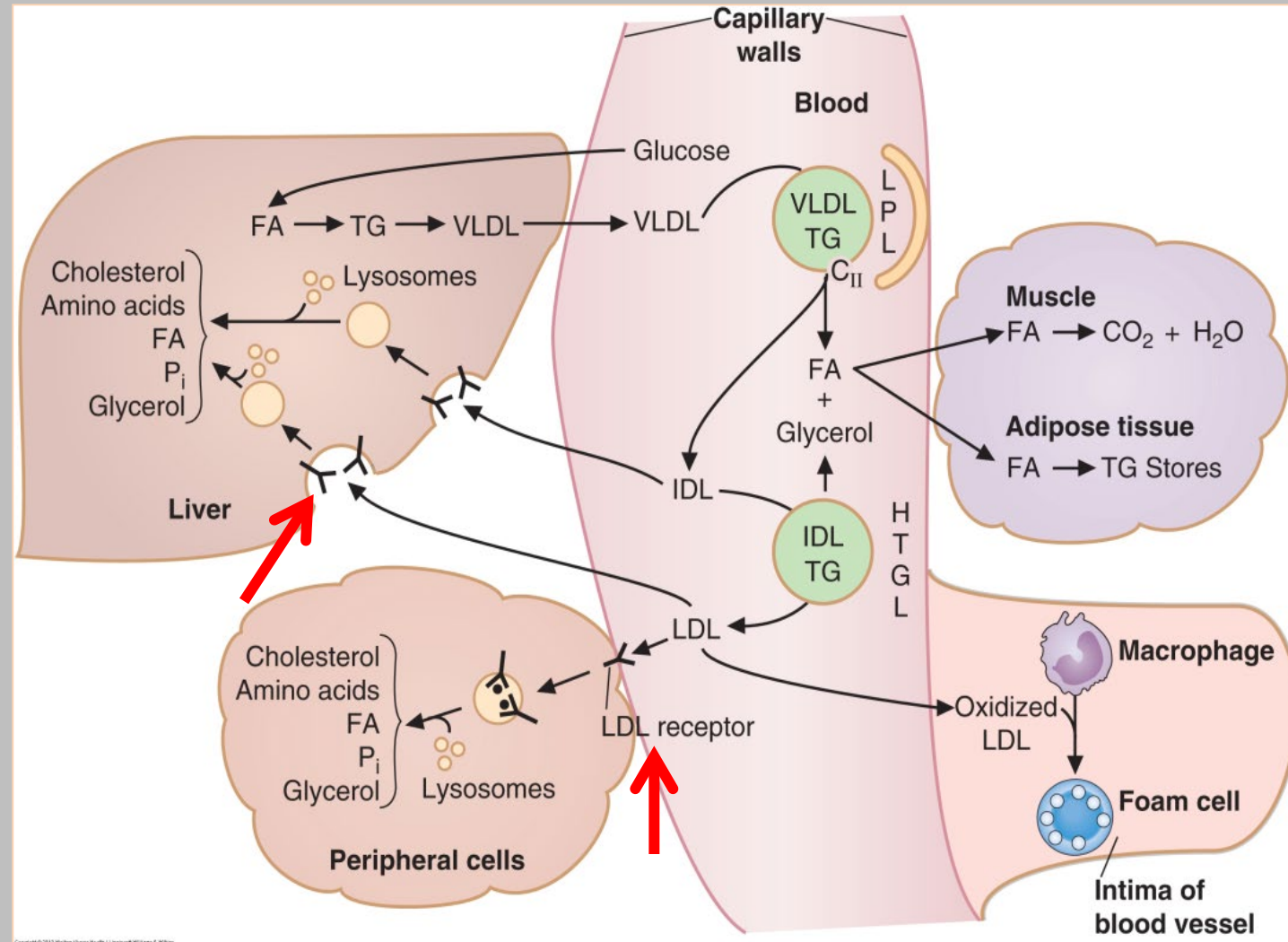
Defective: LDL receptor

ApoB-100

PCSK9

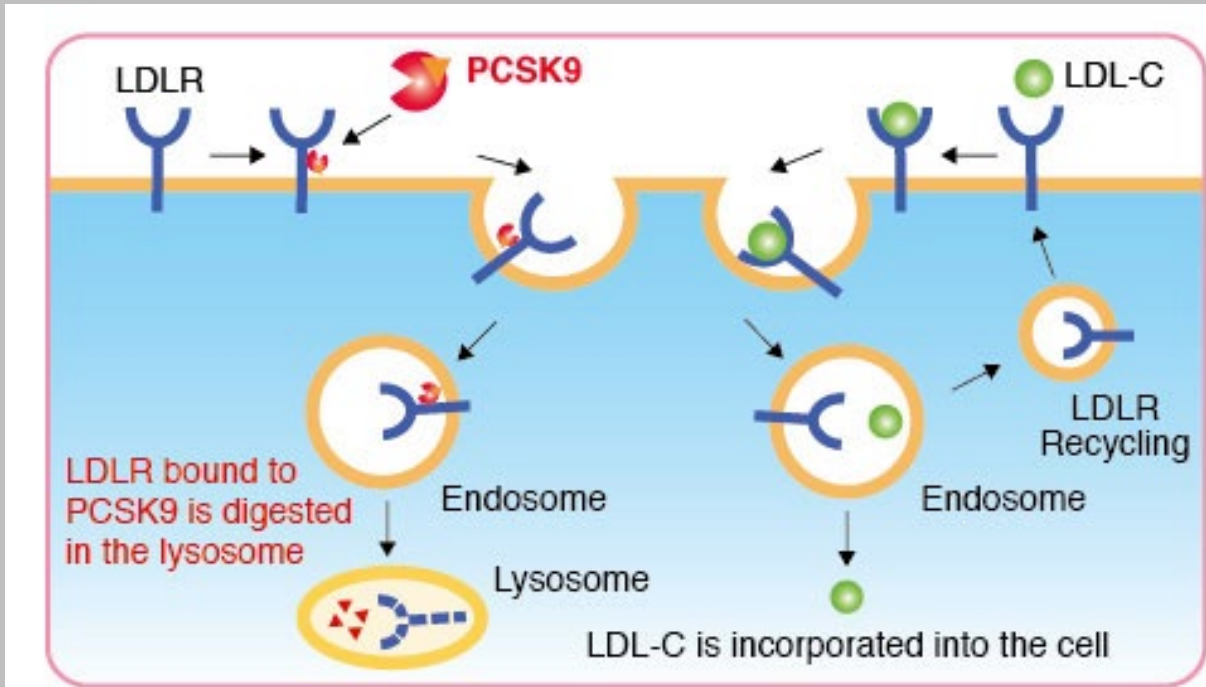
Consequences: LDL ↑, cholesterol ↑.
Increased risk of coronary disease: Yes

- **VLDL** becomes VLDL remnant or IDL after loses **TG**
- **IDL** is further metabolized to become **LDL**
- **LDL** is rich in **cholesterol**
- **LDL** is taken up by liver or peripheral tissues via **ApoB-100** binding to **LDL receptor**
- **PCSK9** regulates the turnover of LDL receptor (next slide)



Treatment of hypercholesterolemia with proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor

- PCSK9 belongs to the proprotein convertase family of serine proteases
- Secreted PCSK9 binds to LDL receptors (LDLR) and targets them for lysosomal degradation
- PCSK9 inhibition $\rightarrow$ LDLR $\uparrow$ $\rightarrow$ LDL $\downarrow$ $\rightarrow$ LDL-Cholesterol $\downarrow$
- PCSK9 inhibitors: Alirocumab (Praluent, Sanofi/Regeneron) and Evolocumab (Repatha, Amgen) are human monoclonal antibodies that inhibit the degradation of LDLR in hepatocytes and thus enhance the uptake and clearance of circulating LDL particles.



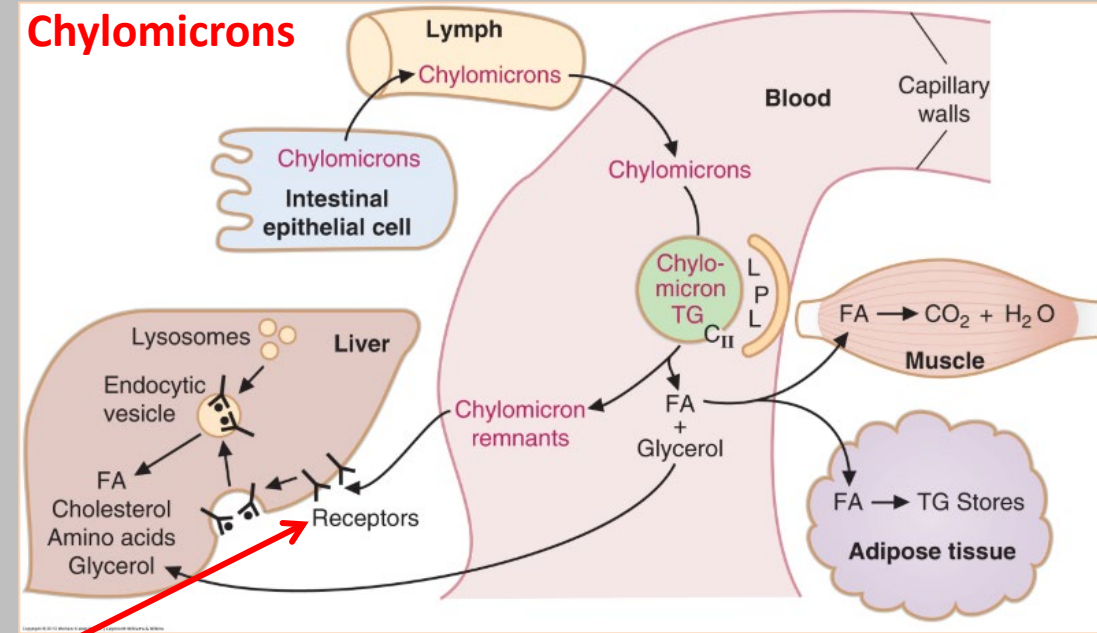
Hyperlipoproteinemia – familial Type III

Defective: ApoE

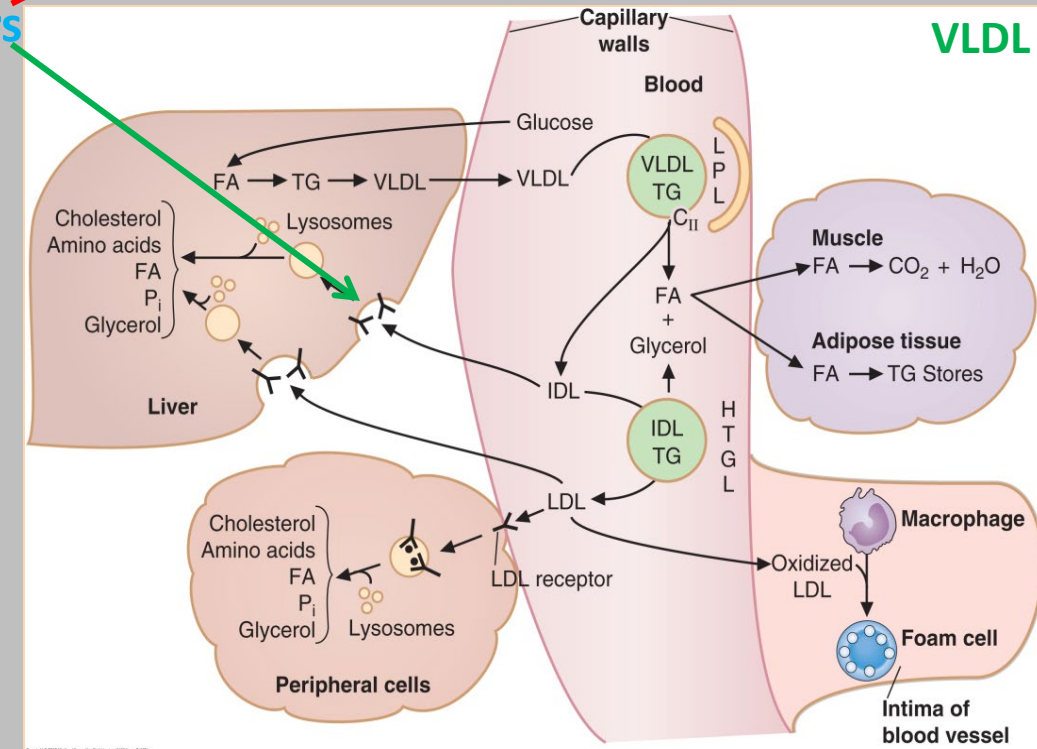
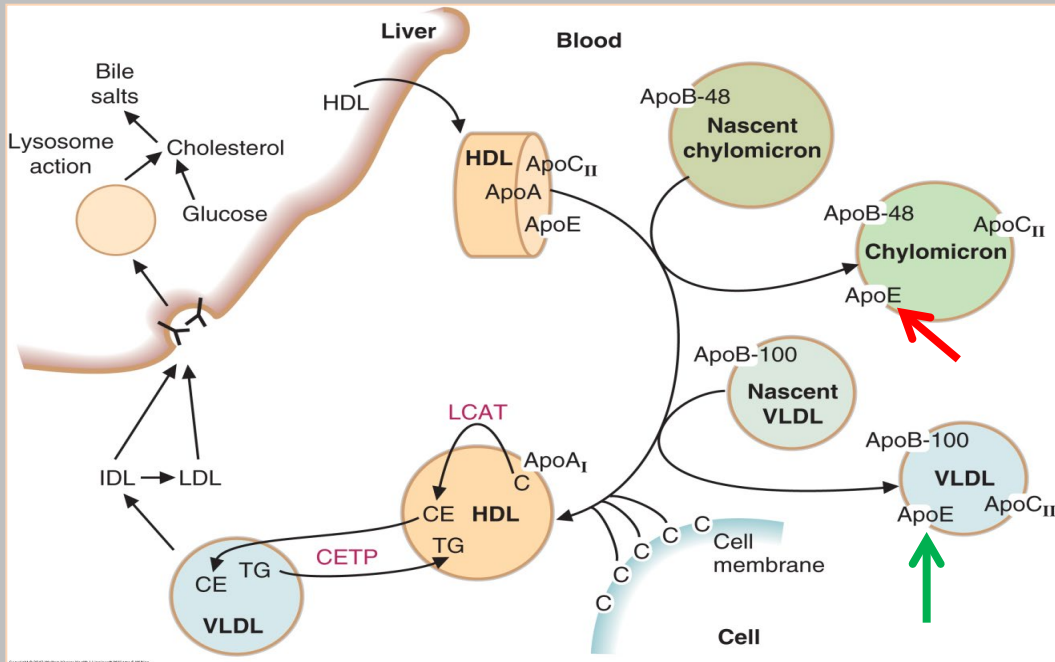
Consequences: chylomicron↑, VLDL↑

Increased risk of coronary disease: Yes

Other clinical manifestation: xanthomas



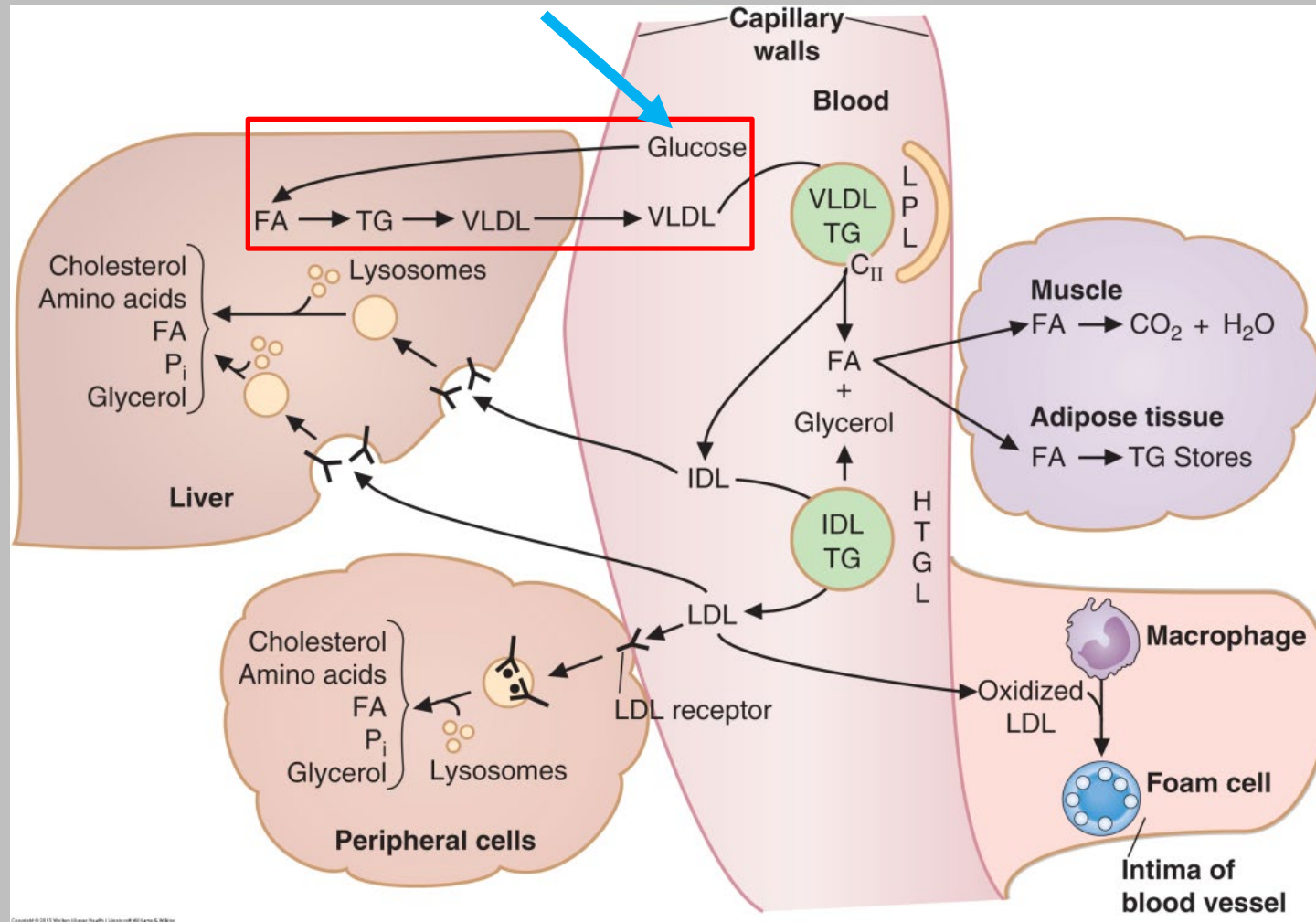
ApoE receptors



Hyperlipoproteinemia (Type IV)

– familial hypertriacylglycerolemia

Overproduction of VLDL often associated with **glucose intolerance** → VLDL ↑, TG ↑.
Commonly associated with coronary heart disease, type 2 diabetes, obesity, etc.



Hypolipoproteinemia

Hypo-: lower than normal



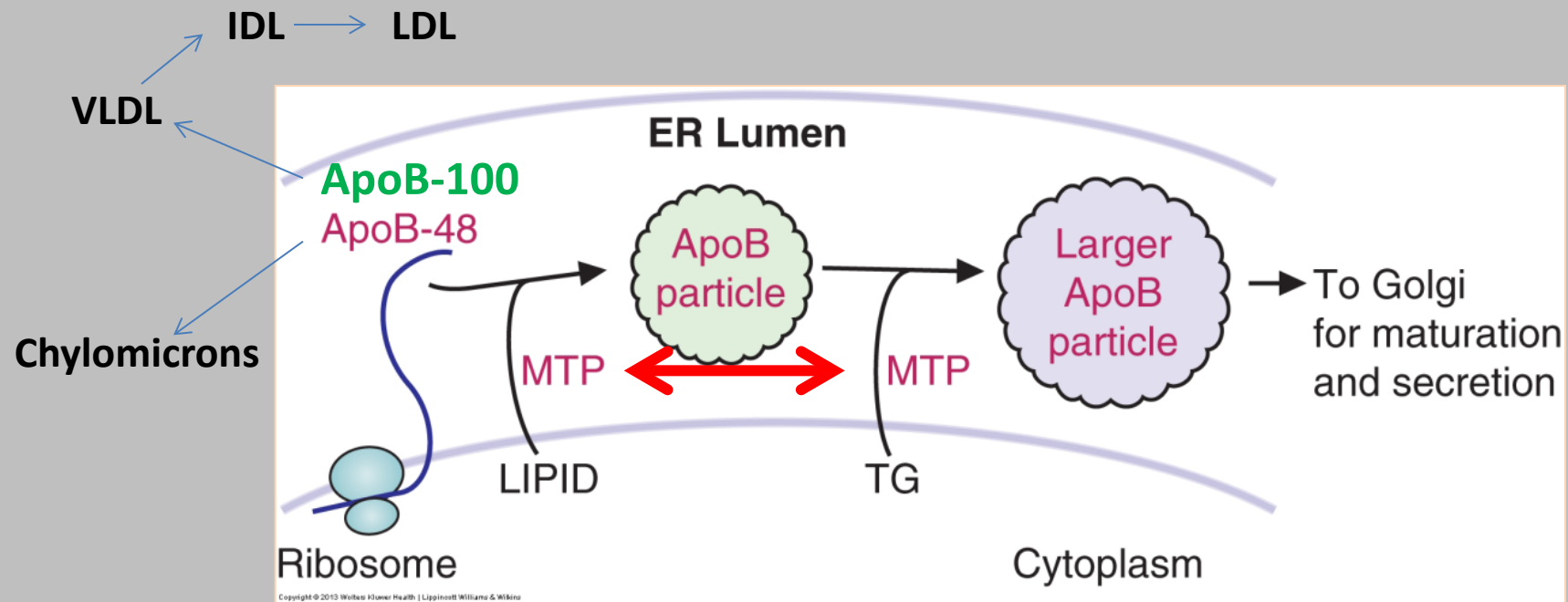
Summary of **hypo**lipoproteinemia

Disorder Type	Cause	Effects
Tangier disease	Defects in ABCA1 , the transporter that supports cholesterol pickup by nascent HDLs	HDL: ↓
A beta lipoproteinemia (Beta: Apo B -containing lipoproteins)	Mutations in the MTTP* (microsomal triglyceride transfer protein, MTP) gene	Chylomicrons: ↓ VLDL, IDL, LDL: ↓
Hypo a lipoproteinemia (Alpha: Apo A -containing lipoproteins)	Accelerated catabolism of ApoA-I and ApoA-II	HDL: ↓

*: The gene *MTTP* encodes the protein MTP.

Hypolipoproteinemia - Abetalipoproteinemia

Disorder Type	Cause	Effects
Abetalipoproteinemia (Beta: ApoB-containing lipoproteins)	Mutations in the MTTP* (microsomal triglyceride transfer protein) gene	Chylomicrons: ↓ VLDL, IDL, LDL: ↓



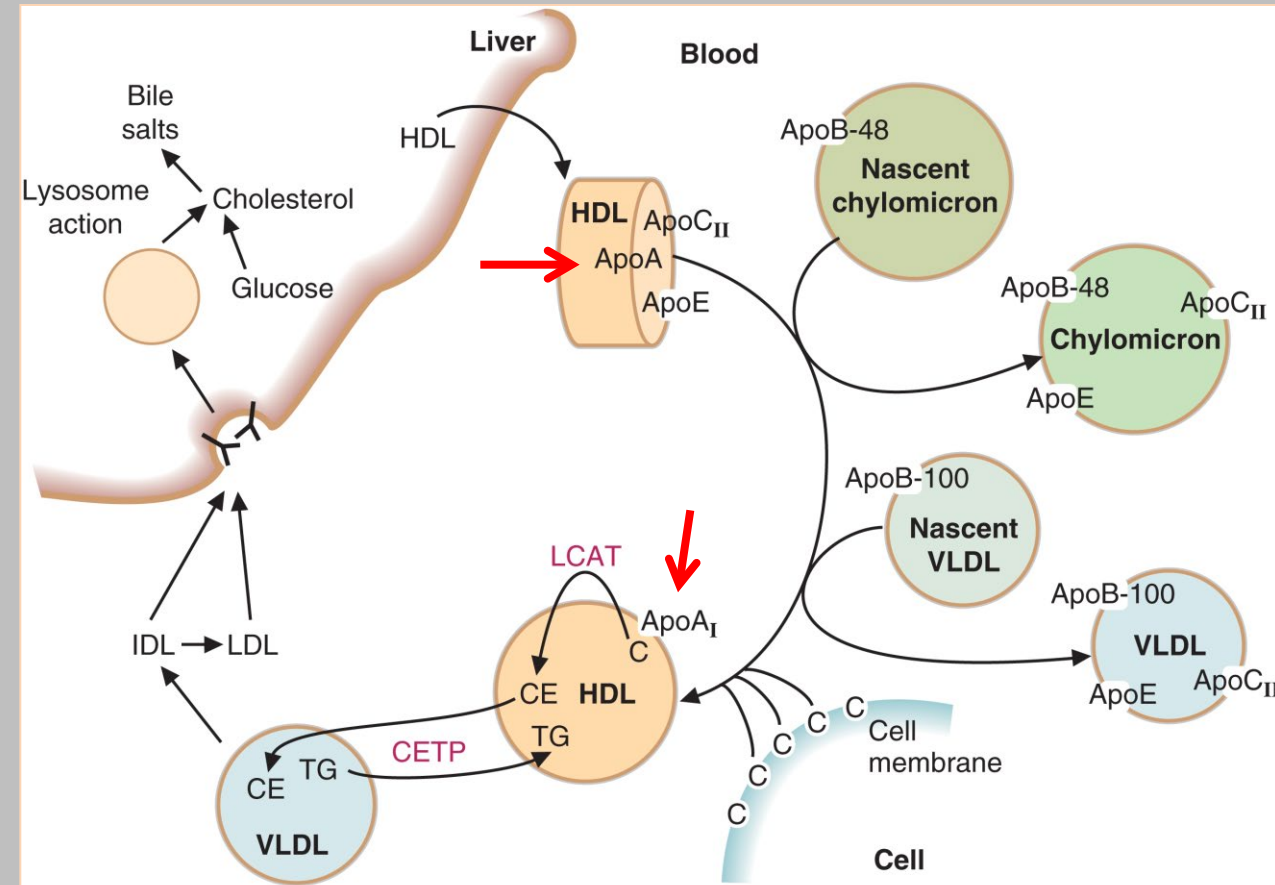
*: The gene *MTTP* encodes the protein MTP.

Hypolipoproteinemia - Hypo α lipoproteinemia

Disorder Type	Cause	Effects
Hypoαlipoproteinemia (α : ApoA-containing lipoproteins)	Accelerated catabolism of ApoA-I and ApoA-II	HDL: ↓

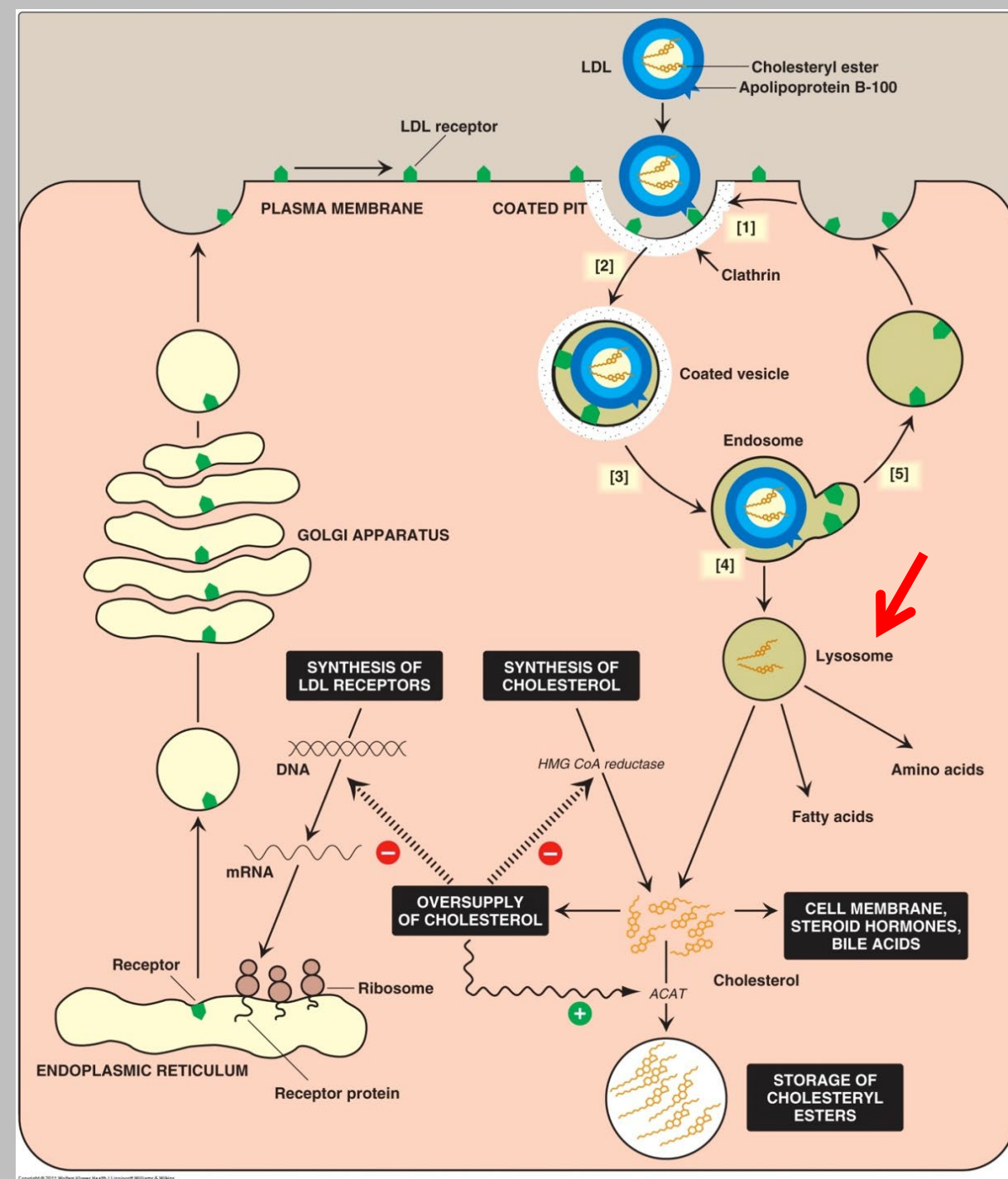
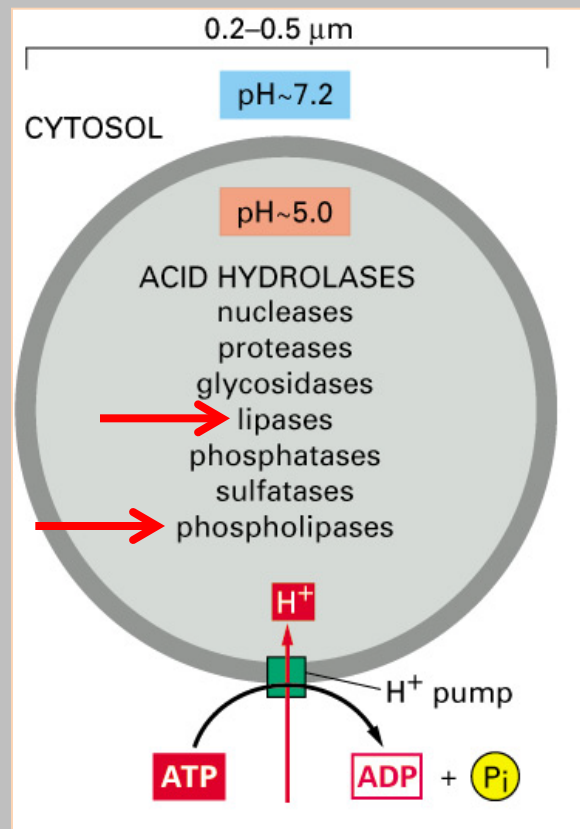
HDL:

- Synthesized mainly in the liver and intestine
- Nascent HDL particle is disk-shaped
- Contains/transfers several apoproteins (**ApoA**, ApoE, and ApoC-II; others)
- **ApoA-I** activates LCAT
- Matures after receiving TG from VLDL and cholesterol from peripheral tissues



Additional genetic diseases related to lipoproteins – **Wolman Disease**

Wolman Disease: **lysosomal acid lipases** hydrolyze **cholesterol esters** and **TGs** within the cell. If these lipases are defective, as in **Wolman disease**, the cholesterol esters and TGs accumulate in the cells and cannot be mobilized when needed, resulting in early death.

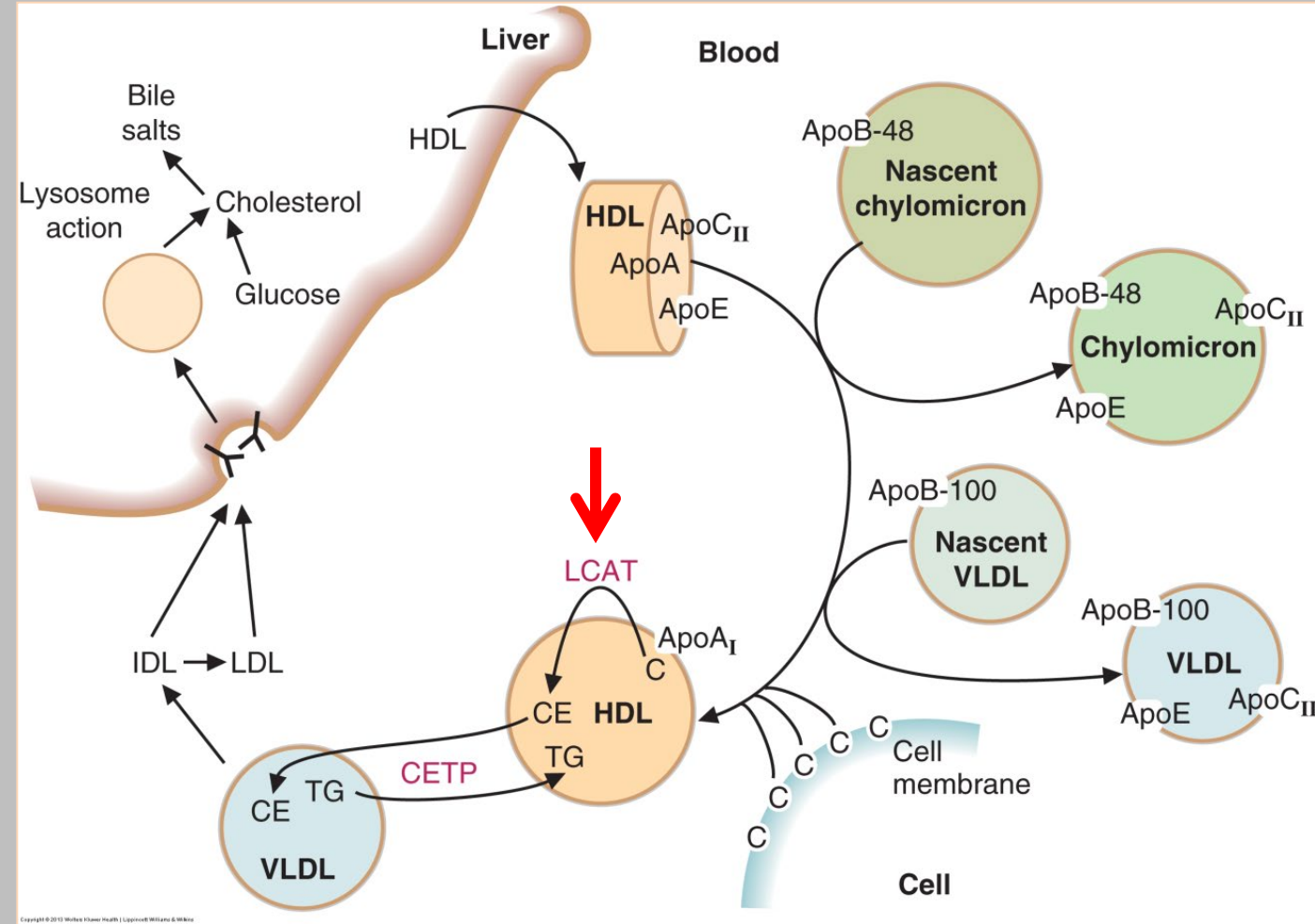




Additional genetic diseases related to lipoproteins– **LCAT deficiency**

LCAT deficiency: LCAT (lecithin - cholesterol acyl transferase) converts **cholesterol** to **cholesterol esters** in HDL.

LCAT deficiency leads to **low HDL levels** and **tissue accumulation of cholesterol**, resulting in anemia, corneal opacification, and renal failure.





Fatty Liver:

If the liver does not have adequate energy available for protein synthesis, or is starved of amino acids, or choline, this will result in **decreased synthesis of apolipoproteins or phosphatidyl choline (lecithin)**, which are necessary for the synthesis of VLDL, then the liver **cannot make VLDL**; so, most of the **TGs** that normally are exported to peripheral tissues through VLDL will not be exported, instead will accumulate in the liver and cause the **fatty liver**.



Clinical goals for cholesterol levels - HDL

- High **serum cholesterol** levels are strongly correlated with cardiovascular disease
- Higher levels of **HDL** are cardio-protective, possibly through both:
 - prevention of oxidative damage to LDL
 - reverse cholesterol transport

Control of serum cholesterol levels		HDL	
<u>Range</u>	<u>Risk</u>	<u>Range</u>	<u>Risk</u>
<200 mg/dL	low	40 mg/dL and lower	high
200 – 239 mg/dL	borderline high	60 mg/dL and higher	low
240 and up mg/dL	high		

Clinical goals for cholesterol levels - LDL

Circulating LDL can suffer **oxidative damage**; such damaged LDL are taken up by **macrophages**, leading to the formation of **foam cells**. Foam cells can accumulate in vasculatures and cause damage to them.

- LDL-cholesterol
 - Goal **<100 mg/dL** (Applies to most individuals who do not have cardiovascular disease or other risk factors for ASCVD)
 - Recommend lifestyle change at **>100 mg/dL**
 - Prescribe medications according to the ACC/AHA guidelines:
(<https://www.acc.org/~media/Non-Clinical/Files-PDFs-Excel-MS-Word-etc/Guidelines/2018/Guidelines-Made-Simple-Tool-2018-Cholesterol.pdf>)

Atherosclerosis: a disease in which plaque builds up inside the arteries.

Different layers of the arterial walls

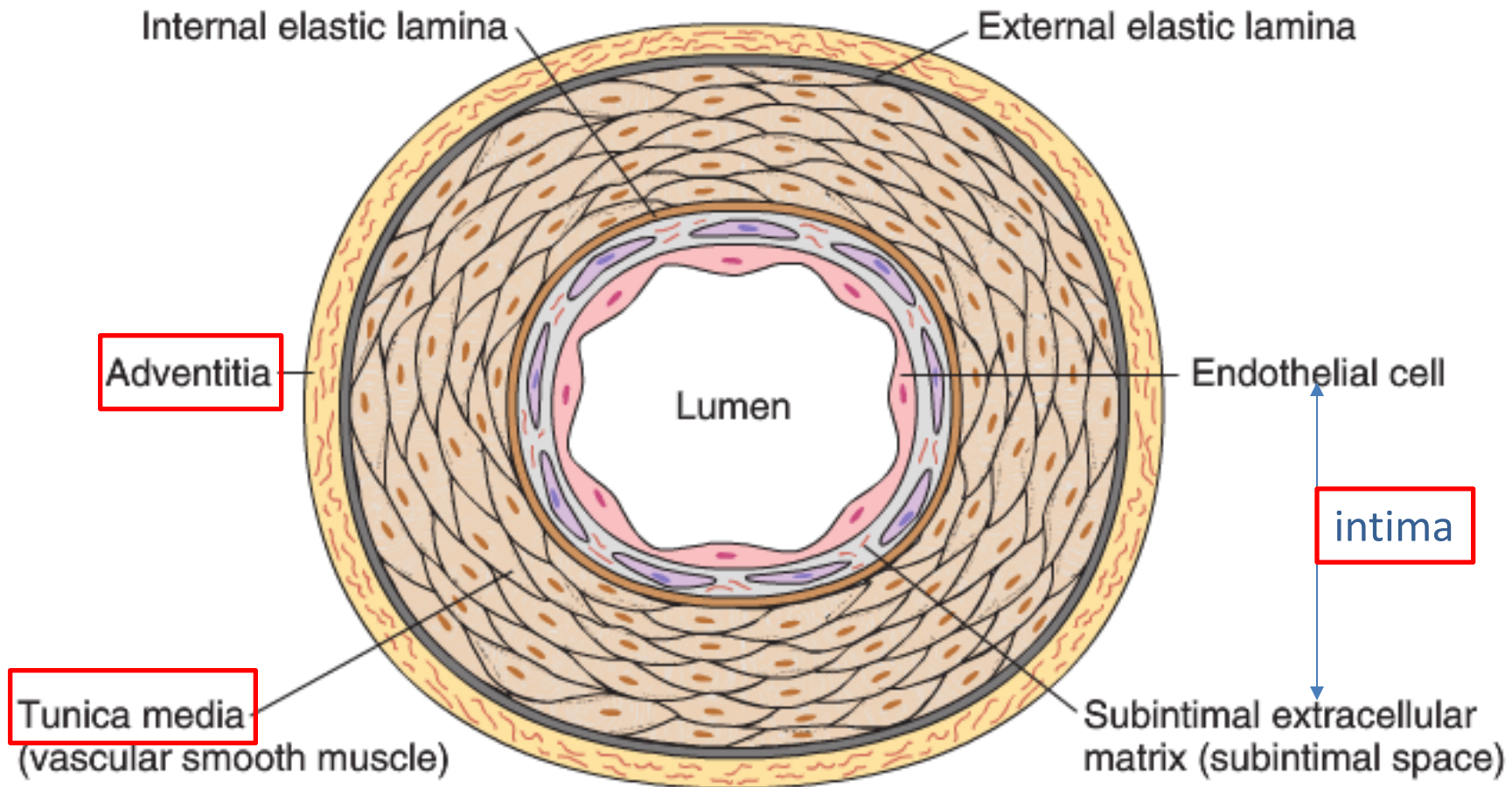
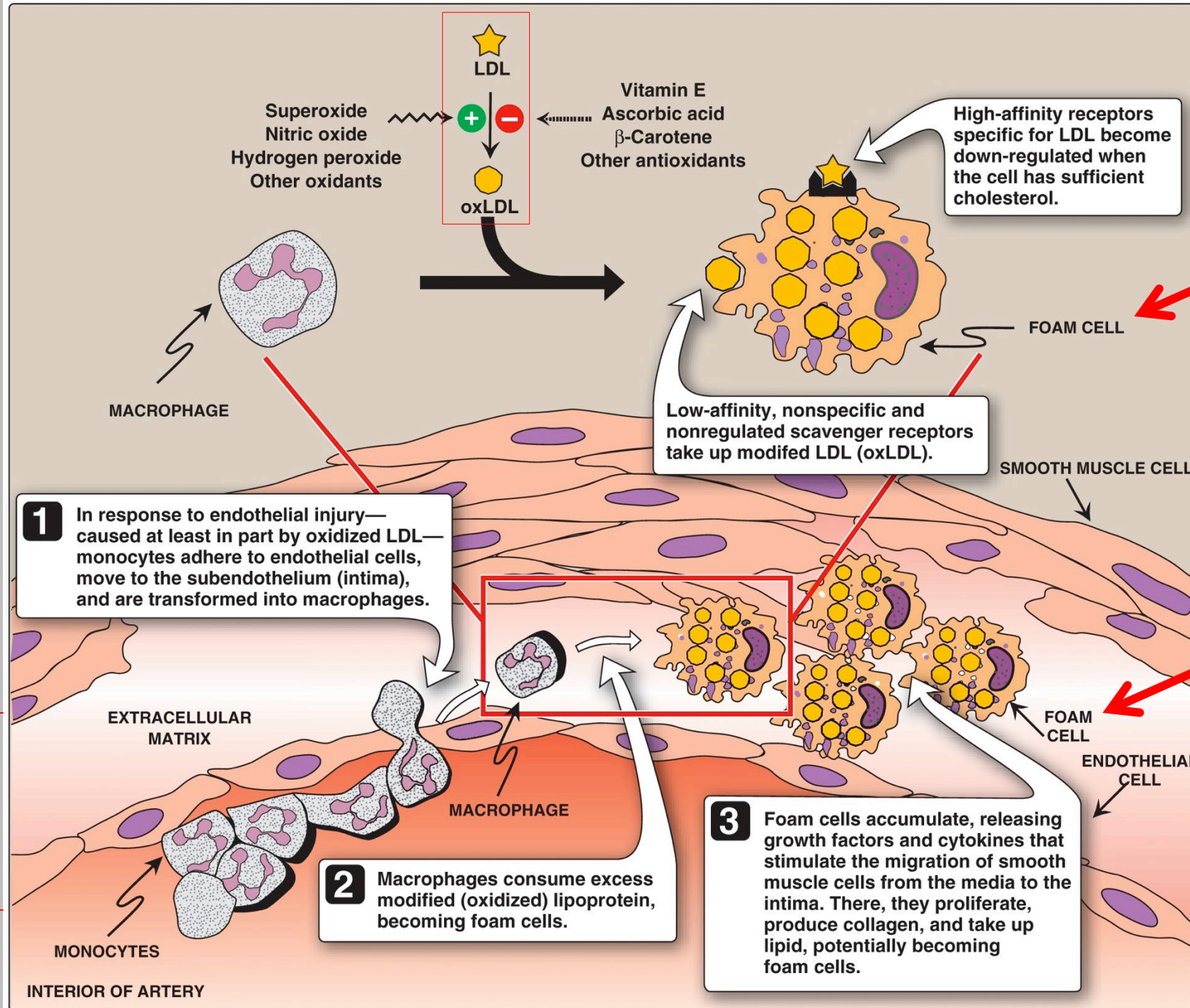


FIG. 34.21. The different layers of the arterial wall.

Intima



The evolution of atherosclerotic plaques

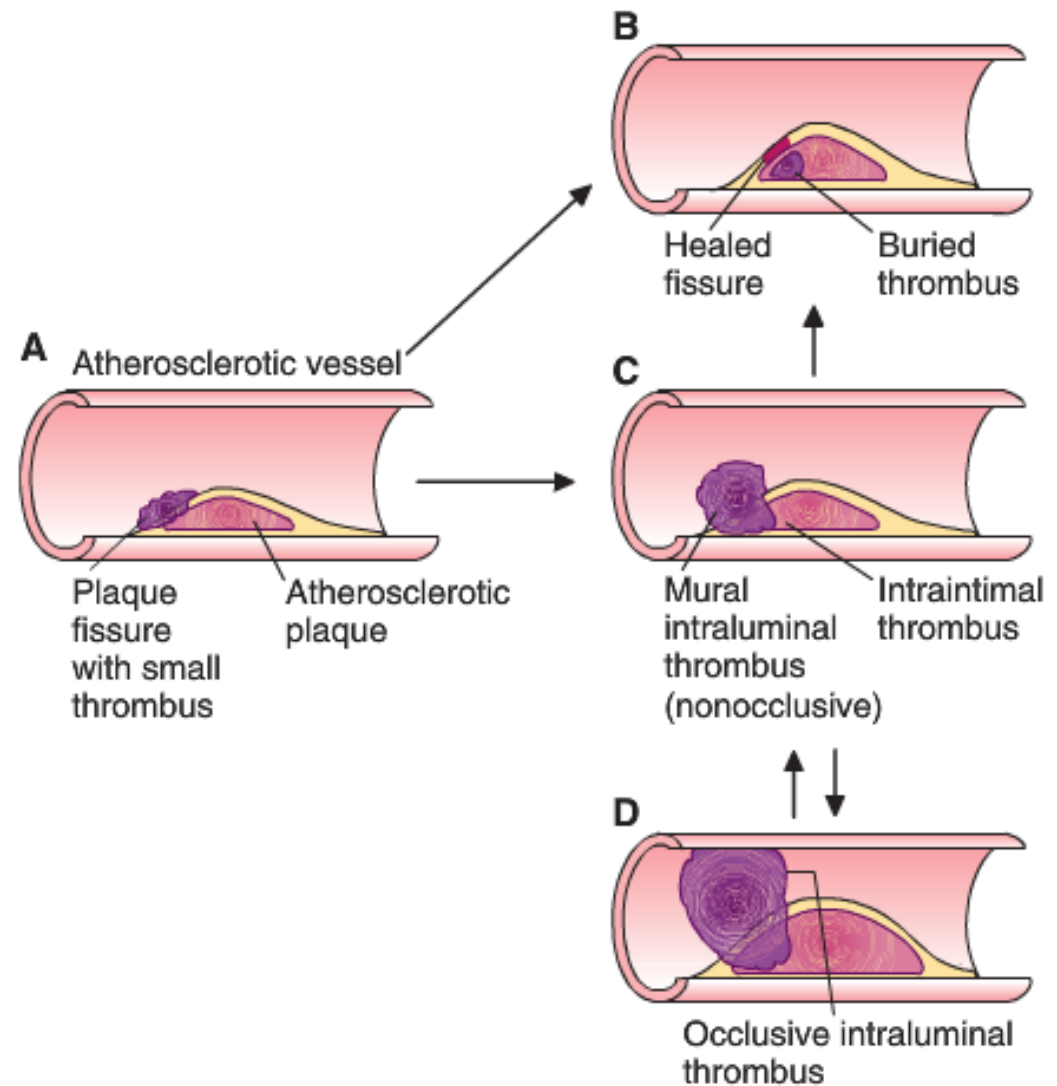


FIG. 34.22. Evolution of an atherosclerotic plaque. Plaque capsule eroded near the “elbow” of plaque creating an early plaque fissure (A), which may heal as plaque increases in size (B) or may grow as thrombus expands, having an intraluminal portion and an intraintimal portion (C). If the fissure is not properly sealed, the thrombus may grow and completely occlude the vessel lumen (D), causing an acute infarction of tissues downstream of the vessel occlusion.

Control of serum cholesterol levels – lifestyle/diet

- Lifestyle/diet modification** used to lower serum cholesterol
- reduced dietary intake of **cholesterol** and **saturated fats**
 - Increased **plant stanols or sterols** (soy)
 - increased **dietary soluble fiber**
 - weight reduction/exercise potentiates drug effects

Table 34.2 Dietary Therapy for Elevated Blood Cholesterol

Nutrient	Step I Diet	Step II Diet^a
Cholesterol ^b	<300 mg/d	<200 mg/d
Total fat	≤30% ^b	30%
Saturated fat	8%–10%	<7%
Polyunsaturated fat	≤10%	≤10%
Monounsaturated fat	≤15%	≤15%
Carbohydrates	≥55%	≥55%
Protein	~15%	~15%
Calories	To achieve and maintain desirable body weight	

^aThe step II diet is applied if 3 months on the step I diet has failed to reduce blood cholesterol to the desired level (see Table 34.1).

^bExcept for the values given in milligram per day, all values are percentage of total calories eaten daily.

Source: Based on National Cholesterol Education Program. Summary of the second report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel II). *JAMA*. 1993;269(23):3015–3023.



Control of serum cholesterol levels - Drugs

- ❖ **HMG CoA reductase inhibitors** (statins: Lipitor, Zocor, Crestor etc.)
- ❖ **PCSK9 inhibitors (Alirocumab and Evolocumab)**: slow LDL receptor degradation
- ❖ **Bile acid sequestrant (Colestipol, Cholestyramine)**: increases bile acid loss in feces
- ❖ **Nicotinic acid (niacin, vitamin B3)(Niaspan, Advicor (with lovastatin))**: mechanism of action not completely understood (reduces Lp(a), as well as LDL-cholesterol and TAGs, and raises HDL)
- ❖ **Fibrate (Gemfibrizol, Clofibrate)**
 - inhibits the **hepatic release of lipoproteins** (particularly VLDL)
 - potentiates the action of **lipoprotein lipase (LPL)**
 - Increase **HDL**
- ❖ **Ezetimibe (Zetia)**
 - inhibits **dietary** absorption of cholesterol
 - combined with a statin in Vytorin

Treatment: Pharmacological

Lipid-lowering agents

DRUG	LDL	HDL	TRIGLYCERIDES	MECHANISMS OF ACTION	ADVERSE EFFECTS/PROBLEMS
HMG-CoA reductase inhibitors (eg, lovastatin, pravastatin)	↓↓↓	↑	↓	Inhibit conversion of HMG-CoA to mevalonate, a cholesterol precursor; ↓ mortality in CAD patients	Hepatotoxicity (↑ LFTs), myopathy (esp. when used with fibrates or niacin)
Bile acid resins Cholestyramine, colestipol, colesevelam	↓↓	Slightly ↑	Slightly ↑	Prevent intestinal reabsorption of bile acids; liver must use cholesterol to make more	GI upset, ↓ absorption of other drugs and fat-soluble vitamins
Ezetimibe	↓↓	↑/—	↓/—	Prevent cholesterol absorption at small intestine brush border	Rare ↑ LFTs, diarrhea
Fibrates Gemfibrozil, bezafibrate, fenofibrate	↓	↑	↓↓↓	Upregulate LPL → ↑ TG clearance Activates PPAR- α to induce HDL synthesis	Myopathy (↑ risk with statins), cholesterol gallstones
Niacin (vitamin B₃)	↓↓	↑↑	↓	Inhibits lipolysis (hormone-sensitive lipase) in adipose tissue; reduces hepatic VLDL synthesis	Red, flushed face, which is ↓ by NSAIDs or long-term use Hyperglycemia Hyperuricemia
PCSK9 inhibitors Alirocumab, evolocumab	↓↓↓	↑	↓	Inactivation of LDL-receptor degradation, increasing amount of LDL removed from bloodstream	Myalgias, delirium, dementia, other neurocognitive effects



Metabolic Syndrome

The **metabolic syndrome** (syndrome X, insulin resistance syndrome): consists of a constellation of **metabolic abnormalities** that confer increased risk of cardiovascular disease (**CVD**) and **diabetes mellitus**.

Risk factors:

- A large waistline: waist circumference >102 cm (M), >88 cm (F)
- A high triglyceride level: ≥ 150 mg/dL
- Low HDL-cholesterol: <40 mg/dL, or <50 mg/dL for men and women, respectively
- High blood pressure: blood pressure ≥ 130 mmHg systolic ,or ≥ 85 mmHg diastolic
- High fasting plasma glucose level: ≥ 100 mg/dL or previously diagnosed **type 2 diabetes**

Summary

- **Key concepts:** Atherosclerosis, metabolic syndrome
- Drugs used to lower cholesterol and the biochemical basis behind them
- **Clinical correlations:** disease symptoms, biochemical basis, and treatment (Hyperlipoproteinemias, Hypolipoproteinemia, Wolman Disease, LCAT deficiency, Fatty Liver)

Q1: A 19 year old male has entered your clinic with severe chest pains, and you suspect a myocardial infarct. Blood analysis also revealed that this patient has cholesterol levels of 325 mg/dL. You tried dietary control with discouraging results. Next, you prescribe lipitor (statin) and cholestyramine. Explain the biochemical mechanisms of these drugs for lowering blood cholesterol. What vitamin in megadoses is used to lower cholesterol?

Q2: You suspect that this patient's high cholesterol is because of familial hypercholesterolemia (FH). What is the molecular basis for FH? Which lipoprotein would you expect to be elevated?

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- <https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>